

Simultaneous Multibeam Resource Allocation Scheme for Multiple Target Tracking

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Abstract—A colocated multiple-input multiple-output (MIMO) radar system has the ability to address multiple beam information. However, the simultaneous multibeam working mode has two finite working resources: the number of beams and the total transmit power of the multiple beams. In this scenario, a resource allocation strategy for the multibeam working mode with the task of tracking multiple targets is developed in this paper. The basis of our technique is to adjust the number of beams and their directions and the transmit power of each beam through feedback, with the purpose of improving the worst tracking performance among the multiple targets. The *Bayesian Cramér–Rao lower bound* (BCRLB) provides us with a lower bound on the estimated *mean square error* (MSE) of the target state. Hence, it is derived and utilized as an optimization criterion for the resource allocation scheme. We prove that the resulting resource optimization problem is nonconvex but can be reformulated as a set of convex problems. Therefore, optimal solutions can be obtained easily, which greatly aids real-time resource management. Numerical results show that the worst case tracking accuracy can be efficiently improved by the proposed *simultaneous multibeam resource allocation* (SMRA) algorithm.

Index Terms—BCRLB, colocated MIMO, convex optimization, multiple targets, simultaneous multibeam resource allocation (SMRA).

I. INTRODUCTION

A. Background and Motivation

THE multiple-input multiple-output (MIMO) radar system has received considerable attention recently [1]–[3] and is on a path from theory to practical use. In general, MIMO radar can be divided into two types, namely, MIMO radar with separated antennae [2] and colocated MIMO radar [3]. In the first scenario, the transmit antennae are located far apart from one another relative to their distance to the target [2]. This system enjoys a diverse gain that is manifested in metrics such as the probability of missed detections [4] and target localization accuracy [5]. However, there are still many actual difficulties that

prevent separated MIMO radar from being applied in practice. These difficulties include multistatic synchronization and detection [6]. The latter type, in which the transmit and receive antennae are located close to each other relative to the target, might be seen as a promotion of the conventional phased array radar [7]. This system can transmit, via its antennae, multiple probing signals that could be correlated or uncorrelated with one another. Subject to the current technical conditions, the monostatic colocated MIMO radar is the most practical MIMO radar system [6]. The possible advantages of the colocated MIMO radars have provided the motivation to explore their capability in various contexts such as target detection [7], target localization [5], target tracking [8], waveform design [9], [10], and antenna allocation [11].

Due to the unique structure of the colocated MIMO radar, various desired beam patterns can be generated by multiple colocated transmitters [9], [10], [12]. Therefore, colocated MIMO radar can launch multiple beams simultaneously to execute several radar tasks independently. One of the potential applications that has recently been exposed is to track multiple targets in a multibeam mode. The expression ‘multibeam’ refers to a mode of operation in which multiple orthogonal transmit beams are synthesized simultaneously by different probing signals from various transmitters. In this mode, the user can lower the peak power to achieve a low probability of interception while extending the integration time to maintain the system’s sensitivity [13].

Technically speaking, these multiple orthogonal beams are not formulated by different transmit antennae one by one; they are jointly synthesized by multiple colocated transmitters [9], [10], [12]. When the transmit illumination is defocused, the receive beams must be chosen to span the volume of space that is illuminated by the transmitter (see Fig. 1). Typically, this is accomplished by using multiple, highly focused receive beams (collectively spanning the region of transmit illumination) [13]. In this scenario, the signals from multiple targets can be separated by using their spatial diversity.

In practice, the simultaneous multibeam working mode has two finite working resources: (1) *The maximum beam number M* . If we assume that the number of radiating elements is N , then restricted by the degrees of freedom, the system can generate at most $M(M \leq N)$ orthogonal transmit beams simultaneously at each time index; (2) *At any illumination, the total transmit power P_{total} of the multiple beams is constrained*. There exists the fact that the consumption of the power in this technique grows greatly with the number of beams. To avoid the radar transmitter consumption power surpassing the endurable ability of the radar system’s physical equipment, the

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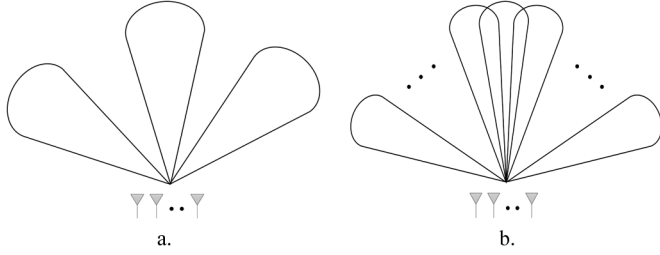


Fig. 1. Simultaneous multiple transmit beams and multiple receive beams. (a) Transmit beam pattern; (b) receive beam pattern.

power constraint must be considered. In this case, how to use the limited resources to achieve the maximal radar potential is of crucial importance in a real application.

In a traditional colocated MIMO radar system, prior knowledge is not used for a resource allocation strategy. For convenience, the number of beams at every instant is fixed, and the predetermined power budget is uniformly allocated to those beams. This arrangement will definitely cause an insufficient use of the limited radar beam and power resources. Fortunately, as a novel approach to enhancing the utilization efficiency of the scarce resource, a cognitive technique can compensate for this defect. A key feature of the expression ‘cognitive’ is to adaptively adjust a radar transmitter to perceive the environment [14]. Hence, with the prior knowledge obtained from the previous recursion cycle in target tracking, it is advantageous for us to optimally and adaptively allocate the limited radar resource.

Several problems have been addressed in the past for resource allocation. In [15], the authors employ a broadband orthogonal frequency division multiplexing signaling at the transmitter and implement adaptive waveform design by minimizing the *Bayesian Cramér-Rao lower bound* (BCRLB) [16] on the target state estimates. Reference [14] employs dynamic programming to select optimal waveforms from a prescribed library using BCRLB as an optimization criterion. In [11], the antenna allocation problem for a colocated MIMO radar system is considered. Moreover, many studies also extend the closed-loop cognitive framework to the netted radar system [17], [18]. Such a cognitive radar network incorporates several radars working together to achieve the task of realizing an enhanced remote sensing capability. For example, [18] computes the BCRLB on the estimates of the target state and the channel state and uses it as an optimization criterion for the antenna selection and power allocation algorithms for in a complex urban environment.

On the basis of the studies mentioned above, we propose a resource allocation strategy for the simultaneous multibeam working mode with the task of tracking multiple, widely spread, independent point targets. We assume that each beam can track only one target for simplicity,¹ and thus, a limited number of targets can be illuminated in this working mode. Our aim is to enhance the utilization efficiency of the scarce radar beam and power resources for the multibeam working mode. Specifically, the resource allocation problems that we want to address are: (1) *How many beams must be generated at every instant;* (2) *Which targets are to be tracked by those beams;* and (3) *How*

¹In our derivation, we also prove that the resource allocation strategy can easily be extended to the case in which multiple targets are covered by a single beam.

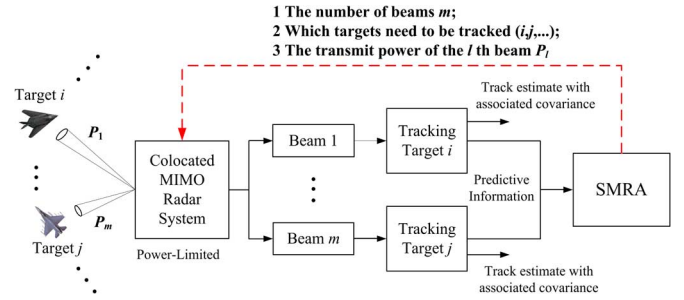


Fig. 2. System model of the SMRA strategy.

much power is to be allocated to each target. The whole algorithm can be viewed as a reaction of the cognitive transmitter to the environment, to improve the worst tracking performance among the multiple targets (we refer it to the worst case tracking performance). The BCRLB provides us with a lower bound on the estimated *mean square error* (MSE) of the target state, and importantly, it can be calculated predictively [16]. Hence, it is derived and utilized as an optimization criterion for the resource allocation scheme. The system model of the *simultaneous multi-beam resource allocation* (SMRA) strategy is given in Fig. 2.

B. Main Contributions

Overall, this paper concentrates on how to design the desired multibeam pattern for certain purposes, while [9], [10] and the references therein focus on how to synthesize the desired multibeam pattern with the use of multiple transmitters. The major contributions of this paper are the following:

- 1) *An SMRA scheme for multiple target tracking (MTT) is proposed.* The BCRLB is derived as a lower bound on the estimates of the target state and channel state. Based on which, the resource allocation algorithm is presented. As such, the specific aim of this paper is to optimally allocate the predetermined beam and power resource, which can result in the minimization of the worst case tracking BCRLB;
- 2) *We build a closed-loop MTT scheme for a colocated MIMO radar system with a simultaneous multibeam working mode.* Due to the unique working mode, the problem of MTT for a number of well-separated targets can be simplified as a number of single-target tracking problems that can be solved independently [19]. Hence, we employ the *sequential importance resampling* particle filter (SIR-PF) [20] technique to obtain an accurate estimate of the hybrid state (target position, velocity and *radar cross section* (RCS)) vector of each target. Then, the predictive information obtained from the tracking recursion cycle is used to form a probabilistic understanding of the environment [21]. Finally, the colocated MIMO radar incorporates this knowledge into its illumination strategy, thereby rendering it to be a closed-loop system;
- 3) *We strictly prove that the nonconvex resource allocation problem can be split into a set of convex problems, and thus, the optimal solution can be obtained easily, which greatly aids real-time applications.*

The remainder of this paper is organized as follows: Section II formulates the system model. In Section III, the *Bayesian information matrix* (BIM), whose inverse yields the BCRLB, is derived. The SMRA scheme is proposed in

Section IV. The basis of the resource allocation scheme is introduced in Section IV-A. Section IV-B derives the criterion of the allocation strategy. The resulting nonconvex optimization problem is reformulated as a set of convex problems and is solved in Section IV-C. In Section IV-D, we divide the MTT problem into a number of single-target tracking problems, and we adopt SIR-PF to achieve the hybrid state estimation for each target. Several numerical results are provided in Section V to verify the effectiveness of the proposed algorithm. Finally, the conclusion of the paper is given in Section VI.

II. SYSTEM MODEL

Consider a monostatic colocated MIMO radar system, which is arbitrarily located at the coordinate (x, y) . A set of Q independent point targets is assumed. The q th target is initially located at (x_{T0}^q, y_{T0}^q) , with an initial speed of $(\dot{x}_{T0}^q, \dot{y}_{T0}^q)$. Set T_0 to be the time interval between successive frames, which we refer to as the tracking interval. The q th target is then located at coordinate (x_{Tk}^q, y_{Tk}^q) at time kT_0 and moves with the speed $(\dot{x}_{Tk}^q, \dot{y}_{Tk}^q)$. The system is to track the locations and has available estimates of some other unknown parameters of these targets, such as the target RCS. This system adopts a multibeam concept in which multiple simultaneous and independent beams are formed, with each of them tracking a distinct target. Restricted by the degrees of freedom, the system can generate at most M ($M \leq Q$) orthogonal transmit beams simultaneously, which implies that at most M targets can be measured at every instant.

To begin with, to simplify the problem, we provide some moderate assumptions:

- 1) The number of targets is known *a priori* (this information can be obtained from the search mode);
- 2) The target is widely distributed in the surveillance region.

A. Signal Model

Assuming that the radar transmits a signal $s_{q,k}(t)$ to the q th target at the k th tracking interval with the following waveform:²

$$s_{q,k}(t) = \sqrt{P_{q,k}} E_{q,k}(t) \exp(-j2\pi f_c t), \quad (1)$$

where f_c is the carrier frequency, and $P_{q,k}$ represents the transmit power, which is defined as

$$\begin{cases} P_{q,k} > 0 & \text{if the } q \text{ th target is tracked} \\ P_{q,k} = 0 & \text{otherwise} \end{cases}. \quad (2)$$

The term $E_{q,k}(t)$ is the normalized complex envelope of the transmit signal, which has an effective bandwidth of $\beta_{q,k}$ [22]

$$\beta_{q,k}^2 = \frac{\left(\int f^2 |E_{q,k}(f)|^2 df \right)}{\left(\int |E_{q,k}(f)|^2 df \right)}, \quad (3)$$

and an effective time duration $T_{q,k}$ [22]

$$T_{q,k}^2 = \frac{\left(\int t^2 |E_{q,k}(t)|^2 dt \right)}{\left(\int |E_{q,k}(t)|^2 dt \right)}, \quad (4)$$

²For brevity, throughout this paper, the target/beam index “ q ” and the time index “ k ” will often be omitted, unless doing so causes confusion.

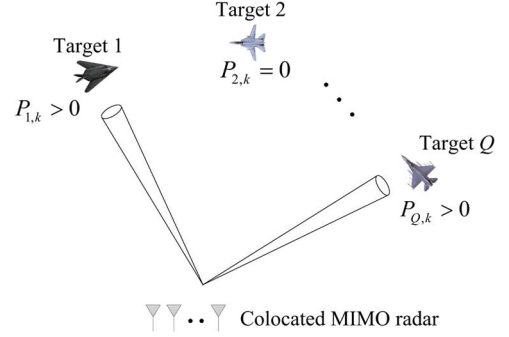


Fig. 3. System model of the colocated MIMO radar system.

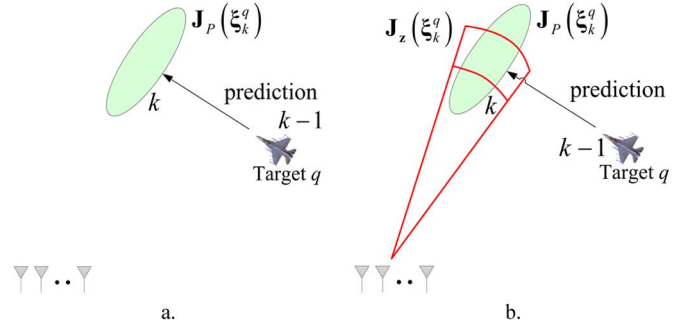


Fig. 4. The relationship between the q th target's predictive BIM and $(u_{q,k}, P_{q,k})$. $\mathbf{J}_z(\xi_k^q)$, whose existence depends on $u_{q,k}$ and size relates to $P_{q,k}$, is the FIM of the data. a) if $u_{q,k} = 0$; b) if $u_{q,k} = 1$.

The sketch map of the colocated MIMO radar system with respect to a set of Q widely distributed targets is shown in Fig. 3. The baseband representation of the received signal is an attenuated version of the transmit signal, which is delayed by

$$r_{q,k}(t) = h_{q,k} \sqrt{\alpha_{q,k} P_{q,k}} E_{q,k}(t - \tau_{q,k}) \exp(-j2\pi f_{q,k} t) + w_{q,k}(t) \quad (5)$$

The attenuation $\alpha_{q,k}$ represents the variation in the signal strength due to path loss effects. In this paper, the target reflectivity $h_{q,k} = h_{q,k}^R + jh_{q,k}^I$ is modeled as a random parameter and must be estimated at every instant. Here, $w_{q,k}(t)$ is a zero-mean, complex white Gaussian noise.

B. Target Dynamic Model

The target motion is prescribed by a *constant-velocity* (CV) model [23]

$$\mathbf{x}_k^q = \mathbf{F}_q \mathbf{x}_{k-1}^q + \boldsymbol{\rho}_{k-1}^q, \quad (6)$$

where the target state is given by $\mathbf{x}_k^q = [x_{Tk}^q, \dot{x}_{Tk}^q, y_{Tk}^q, \dot{y}_{Tk}^q]^T$ with a dimension of $n_x = 4$. Here, $[x_{Tk}^q, y_{Tk}^q]$ and $[\dot{x}_{Tk}^q, \dot{y}_{Tk}^q]$ denote the position and velocity of the q th target, respectively. $\mathbf{F}_q$ is the transition matrix

$$\mathbf{F}_q = \mathbf{I}_2 \otimes \begin{bmatrix} 1 & T_0 \\ 0 & 1 \end{bmatrix}. \quad (7)$$

where $\otimes$ is the Kronecker operator, and $\mathbf{I}_m$ denotes an identity matrix of order m . The term $\boldsymbol{\rho}_{k-1}^q$ in (6) denotes the process

noise and is assumed to be zero-mean Gaussian distributed with a known covariance $\mathbf{Y}_{k-1}^q$ [24]

$$\mathbf{Y}_{k-1}^q = s_q \mathbf{I}_2 \otimes \begin{bmatrix} \frac{1}{3} T_0^3 & \frac{1}{2} T_0^2 \\ \frac{1}{2} T_0^2 & T_0 \end{bmatrix} \quad (8)$$

where s_q denotes the level of the process noise for the q th target.

The transition model of the q th target's channel is assumed to be a first-order Markovian process [17], [18], and it is described by the following equation:

$$\mathbf{h}_k^q = \mathbf{h}_{k-1}^q + \boldsymbol{\mu}_{k-1}^q \quad (9)$$

where the noise $\boldsymbol{\mu}_{k-1}^q$ is white Gaussian with a known covariance $\mathbf{Y}_{\mathbf{h},k-1}^q$. The term $\mathbf{h}_k^q = [h_{q,k}^R, h_{q,k}^I]^T$ represents the channel state vector. Thus, we form an extended state vector for the q th target by concatenating the target state vector and the channel state vector into a single vector of dimension $n_x + 2$, which can be defined as $\boldsymbol{\xi}_k^q = (\mathbf{x}_k^{qT}, \mathbf{h}_k^{qT})^T$. The state transition equation for $\boldsymbol{\xi}_k^q$ is then

$$\boldsymbol{\xi}_k^q = \mathbf{F}_{\boldsymbol{\xi}}^q \boldsymbol{\xi}_{k-1}^q + \boldsymbol{\eta}_{k-1}^q \quad (10)$$

where $\mathbf{F}_{\boldsymbol{\xi}}^q$ is the overall transition matrix

$$\mathbf{F}_{\boldsymbol{\xi}}^q = \begin{pmatrix} \mathbf{F}_q & \mathbf{0}_{n_x \times 2} \\ \mathbf{0}_{2 \times n_x} & \mathbf{I}_2 \end{pmatrix} \quad (11)$$

The term $\boldsymbol{\eta}_{k-1}^q$ is the corresponding Gaussian process noise with a zero mean and covariance of $\mathbf{Y}_{\boldsymbol{\xi},k-1}^q = \text{blkdiag}\{\mathbf{Y}_{k-1}^q, \mathbf{Y}_{\mathbf{h},k-1}^q\}$. Hereafter, when we say state vector, we refer to the extended state vector.

C. Measurement Model

Some elementary measurements can be extracted from the received signal, such as the range, bearing, Doppler shift and target RCS. At the sampling time k , the measurement of the q th target $\mathbf{z}_{q,k}$ has the form

$$\mathbf{z}_{q,k} = \mathbf{g}_{q,k}(\boldsymbol{\xi}_k^q) + \mathbf{w}_{q,k} \quad (12)$$

Hence, the dimension of the measurement is $n_z = 5$, where

$$\mathbf{g}_{q,k}(\cdot) = \begin{bmatrix} g_{R,q,k}(\cdot), g_{\theta,q,k}(\cdot), g_{f,q,k}(\cdot), g_{h_{q,k}^R}(\cdot), g_{h_{q,k}^I}(\cdot) \end{bmatrix}^T \quad (13)$$

with

$$\begin{cases} R_{q,k} = g_{R,q,k}(\boldsymbol{\xi}_k^q) = \sqrt{(x_{T,k}^q - x)^2 + (y_{T,k}^q - y)^2} \\ \theta_{q,k} = g_{\theta,q,k}(\boldsymbol{\xi}_k^q) = \arctan \left[\frac{(y_{T,k}^q - y)}{(x_{T,k}^q - x)} \right] \\ f_{q,k} = g_{f,q,k}(\boldsymbol{\xi}_k^q) = -\frac{2}{\lambda} (\dot{x}_{T,k}^q, \dot{y}_{T,k}^q) \begin{pmatrix} x_{T,k}^q - x \\ y_{T,k}^q - y \end{pmatrix} / R_{q,k} \\ h_{q,k}^R = g_{h_{q,k}^R}(\boldsymbol{\xi}_k^q) = \left(\mathbf{e}_{n_x^q+1}^{n_x^q+2} \right)^T \boldsymbol{\xi}_k^q \\ h_{q,k}^I = g_{h_{q,k}^I}(\boldsymbol{\xi}_k^q) = \left(\mathbf{e}_{n_x^q+2}^{n_x^q+2} \right)^T \boldsymbol{\xi}_k^q \end{cases} \quad (14)$$

which corresponds to different measurement components. In (14), λ denotes the carrier wavelength, and $\mathbf{e}_i^j$ is a zero vector of length j in which the i th element is 1.

We assume that the measurement error $\mathbf{w}_{q,k}$ is a zero-mean Gaussian with a covariance of $\boldsymbol{\Sigma}_{q,k}$:

$$\boldsymbol{\Sigma}_{q,k} = \text{blkdiag} \left(\sigma_{R_{q,k}}^2, \sigma_{\theta_{q,k}}^2, \sigma_{f_{q,k}}^2, \sigma_{h_{q,k}^R}^2, \sigma_{h_{q,k}^I}^2 \right), \quad (15)$$

where $\sigma_{R_{q,k}}^2, \sigma_{\theta_{q,k}}^2, \sigma_{f_{q,k}}^2, \sigma_{h_{q,k}^R}^2$ and $\sigma_{h_{q,k}^I}^2$ are the CRLBs on the estimation MSEs of the q th target's range, bearing, Doppler and RCS [22], [25] information:

$$\begin{cases} \sigma_{R_{q,k}}^2 \propto \left(\alpha_{q,k} P_{q,k} |h_{q,k}|^2 \beta_{q,k}^2 \right)^{-1} \\ \sigma_{\theta_{q,k}}^2 \propto \left(\frac{\alpha_{q,k} P_{q,k} |h_{q,k}|^2}{B_{NN}} \right)^{-1} \\ \sigma_{f_{q,k}}^2 \propto \left(\alpha_{q,k} P_{q,k} |h_{q,k}|^2 T_{q,k}^2 \right)^{-1} \\ \sigma_{h_{q,k}^R}^2 = \sigma_{h_{q,k}^I}^2 \propto (2\alpha_{q,k} P_{q,k})^{-1} \end{cases} \quad (16)$$

where B_{NN} is the null-to-null beam width of the receiver antenna [25].

The general state space model of the q th target is given by (10) and (12). At the k th tracking interval, the parameter to be estimated is the hybrid target state $\boldsymbol{\xi}_k^q$. However, the inaccuracy of the dynamic model and the measurement model could lead to uncertainty in the estimated results. In the next section, the analytical expression of the BIM, which provides a bound on the performance of the estimators of the target state $\boldsymbol{\xi}_k^q$, will be given. Organizing the transmit parameter to control the target state estimation error bounds (determined from the sequence of BIMs), which forms the basis of the SMRA technique, will then be introduced in Section IV.

III. BAYESIAN FISHER INFORMATION MATRIX

Let $\hat{\boldsymbol{\xi}}_{k|k}^q(\mathbf{z}_{q,k})$ be an estimate of $\boldsymbol{\xi}_k^q$, which is a function of the measurement vector $\mathbf{z}_{q,k}$. The Bayesian Cramér-Rao inequality [16] shows that the MSE of any estimator cannot go below a certain bound:

$$\mathbb{E}_{\boldsymbol{\xi}_k^q, \mathbf{z}_{q,k}} \left\{ \left(\hat{\boldsymbol{\xi}}_{k|k}^q(\mathbf{z}_{q,k}) - \boldsymbol{\xi}_k^q \right) \left(\hat{\boldsymbol{\xi}}_{k|k}^q(\mathbf{z}_{q,k}) - \boldsymbol{\xi}_k^q \right)^T \right\} \geq \mathbf{J}^{-1}(\boldsymbol{\xi}_k^q) \quad (17)$$

where $\mathbb{E}_{\boldsymbol{\xi}_k^q, \mathbf{z}_{q,k}}$ denotes the mathematical expectation that is taken with respect to the target state, and the measurement. $\mathbf{J}(\boldsymbol{\xi}_k^q)$ is the BIM of the q th target's state

$$\mathbf{J}(\boldsymbol{\xi}_k^q) = -\mathbb{E}_{\boldsymbol{\xi}_k^q, \mathbf{z}_{q,k}} \left[\Delta_{\boldsymbol{\xi}_k^q}^{\boldsymbol{\xi}_k^q} \ln p(\mathbf{z}_{q,k}, \boldsymbol{\xi}_k^q) \right]. \quad (18)$$

where the notation $\Delta_{\boldsymbol{\xi}_k^q}^{\boldsymbol{\xi}_k^q} = \Delta_{\boldsymbol{\eta}} \Delta_{\boldsymbol{\kappa}}^T$ represents the second-order partial derivative vectors.

The $p(\mathbf{z}_{q,k}, \boldsymbol{\xi}_k^q)$ in (18) is the joint PDF of $(\mathbf{z}_{q,k}, \boldsymbol{\xi}_k^q)$, which can be factorized as

$$p(\mathbf{z}_{q,k}, \boldsymbol{\xi}_k^q) = p(\boldsymbol{\xi}_k^q) p(\mathbf{z}_{q,k} | \boldsymbol{\xi}_k^q), \quad (19)$$

where $p(\boldsymbol{\xi}_k^q)$ is the PDF of the target state, and $p(\mathbf{z}_{q,k} | \boldsymbol{\xi}_k^q)$ is the joint conditional PDF

$$p(\mathbf{z}_{q,k} | \boldsymbol{\xi}_k^q) = \mathcal{N}(\mathbf{g}_{q,k}(\boldsymbol{\xi}_k^q), \boldsymbol{\Sigma}_{q,k}). \quad (20)$$

where $\mathcal{N}(\mathbf{m}, \mathbf{C})$ is a Gaussian density with a mean $\mathbf{m}$ and covariance $\mathbf{C}$. According to (16), if $P_{q,k} = 0$ (which means that there is no measurement), then $p(\mathbf{z}_{q,k} | \boldsymbol{\xi}_k^q)$ is actually uniformly distributed.

Reference [16] provides an elegant method of computing the BIM $\mathbf{J}(\xi_k^q)$ recursively without manipulating the large matrices at each time index:

$$\mathbf{J}(\xi_k^q) = \mathbf{J}_P(\xi_k^q) + \mathbf{J}_Z(\xi_k^q), \quad (21)$$

where $\mathbf{J}_P(\xi_k^q)$ and $\mathbf{J}_Z(\xi_k^q)$ are the *Fisher information matrix* (FIM) of the prior information and the data, respectively

$$\begin{cases} \mathbf{J}_P(\xi_k^q) = -\mathbb{E}_{\xi_k^q} \left\{ \Delta_{\xi_k^q}^{\xi_k^q} \ln p(\xi_k^q) \right\} \\ \quad = (\mathbf{D}_{k-1}^{22} - \mathbf{D}_{k-1}^{21} (\mathbf{J}(\xi_{k-1}^q) + \mathbf{D}_{k-1}^{11})^{-1} \mathbf{D}_{k-1}^{12}) \\ \mathbf{J}_Z(\xi_k^q) = -\mathbb{E}_{\xi_k^q, \mathbf{z}_{q,k}} \left\{ \Delta_{\xi_k^q}^{\xi_k^q} \ln p(\mathbf{z}_{q,k} | \xi_k^q) \right\} \end{cases} \quad (22)$$

where

$$\begin{cases} \mathbf{D}_{k-1}^{11} = -\mathbb{E}_{\xi_{k-1}^q, \xi_k^q} \left\{ \Delta_{\xi_{k-1}^q}^{\xi_{k-1}^q} \ln p(\xi_{k-1}^q | \xi_k^q) \right\} \\ \mathbf{D}_{k-1}^{12} = -\mathbb{E}_{\xi_{k-1}^q, \xi_k^q} \left\{ \Delta_{\xi_{k-1}^q}^{\xi_{k-1}^q} \ln p(\xi_{k-1}^q | \xi_k^q) \right\} \\ \mathbf{D}_{k-1}^{22} = -\mathbb{E}_{\xi_{k-1}^q, \xi_k^q} \left\{ \Delta_{\xi_{k-1}^q}^{\xi_{k-1}^q} \ln p(\xi_{k-1}^q | \xi_k^q) \right\} \end{cases} = (\mathbf{D}_{k-1}^{21})^T. \quad (23)$$

For the linear Gaussian case described in (10), the terms $\mathbf{D}_{k-1}^{11}$, $\mathbf{D}_{k-1}^{12}$ and $\mathbf{D}_{k-1}^{22}$ are deterministic, and the expectation operator can be dropped out

$$\begin{cases} \mathbf{D}_{k-1}^{11} = (\mathbf{F}_{\xi}^q)^T (\mathbf{Y}_{\xi, k-1}^q)^{-1} \mathbf{F}_{\xi}^q \\ \mathbf{D}_{k-1}^{12} = -(\mathbf{F}_{\xi}^q)^T (\mathbf{Y}_{\xi, k-1}^q)^{-1} \\ \mathbf{D}_{k-1}^{22} = (\mathbf{Y}_{\xi, k-1}^q)^{-1} \end{cases}. \quad (24)$$

Substituting (24) into the first equation of (22) and using the matrix inversion lemma as [23], the FIM of the prior information $\mathbf{J}_P(\xi_k^q)$ can be calculated as

$$\mathbf{J}_P(\xi_k^q) = \left[\mathbf{Y}_{\xi, k-1}^q + \mathbf{F}_{\xi}^q \mathbf{J}^{-1}(\xi_{k-1}^q) (\mathbf{F}_{\xi}^q)^T \right]^{-1}. \quad (25)$$

The FIM of the data $\mathbf{J}_Z(\xi_k^q)$ can be evaluated by substituting the joint conditional PDF given in (20) into (22)

$$\begin{aligned} \mathbf{J}_Z(\xi_k^q) &= -\mathbb{E}_{\xi_k^q, \mathbf{z}_{q,k}} \left\{ \Delta_{\xi_k^q}^{\xi_k^q} \ln p(\mathbf{z}_{q,k} | \xi_k^q) \right\} \\ &= -\mathbb{E}_{\xi_k^q} \left\{ \mathbb{E}_{\mathbf{z}_{q,k} | \xi_k^q} \left\{ \Delta_{\xi_k^q}^{\xi_k^q} \ln p(\mathbf{z}_{q,k} | \xi_k^q) \right\} \right\} \\ &= \mathbb{E}_{\xi_k^q} \left\{ \mathbf{H}_{q,k}^T \Sigma_{q,k}^{-1} \mathbf{H}_{q,k} \right\} \end{aligned} \quad (26)$$

where $\mathbf{H}_{q,k} = [\Delta_{\xi_k^q}^{\mathbf{g}_{q,k}}(\xi_k^q)]^T$ is the Jacobian matrix of the measurement function $\mathbf{g}_{q,k}(\xi_k^q)$ with respect to the target state ξ_k^q . Finally, we have the BIM

$$\begin{aligned} \mathbf{J}(\xi_k^q) &= \left[\mathbf{Y}_{\xi, k-1}^q + \mathbf{F}_{\xi}^q \mathbf{J}^{-1}(\xi_{k-1}^q) (\mathbf{F}_{\xi}^q)^T \right]^{-1} \\ &\quad + \mathbb{E}_{\xi_k^q} \left\{ \mathbf{H}_{q,k}^T \Sigma_{q,k}^{-1} \mathbf{H}_{q,k} \right\} \end{aligned} \quad (27)$$

We note that the BIM $\mathbf{J}(\xi_k^q)$ depends on $P_{q,k}$ through the measurement covariance $\Sigma_{q,k}$. From (16), we know that if $P_{q,k} = 0$,

then $\Sigma_{q,k}^{-1}$ will be an all-zero matrix. Hence, to simplify the later derivation, we introduce a vector of binary variables

$$u_{q,k} = \begin{cases} 1, & \text{if } P_{q,k} > 0 \\ 0, & \text{if } P_{q,k} = 0 \end{cases} \quad (28)$$

The BIM can then be reformulated as

$$\begin{aligned} \mathbf{J}(\xi_k^q) &= \left[\mathbf{Y}_{\xi, k-1}^q + \mathbf{F}_{\xi}^q \mathbf{J}^{-1}(\xi_{k-1}^q) (\mathbf{F}_{\xi}^q)^T \right]^{-1} \\ &\quad + u_{q,k} \mathbb{E}_{\xi_k^q} \left\{ \mathbf{H}_{q,k}^T \Sigma_{q,k}^{-1} \mathbf{H}_{q,k} \right\} \\ &= \mathbf{J}_P(\xi_k^q) + u_{q,k} \mathbf{J}_Z(\xi_k^q) \end{aligned} \quad (29)$$

In practice, the second term of the right-hand side of (29), which denotes the FIM of the data matrix, will be evaluated using Monte Carlo techniques [23]. For the multi-target case, we must implement the Monte Carlo simulation M times to achieve the BIM, which could cost a substantial amount of time. Implementing the resource allocation techniques into real-time systems necessitates a quick and efficient calculation of the BCRLB. To achieve this goal, we make a small process noise assumption and approximate the BIM as follows [26]

$$\mathbf{J}(\xi_k^q) = \mathbf{J}_P(\xi_k^q) + u_{q,k} \hat{\mathbf{H}}_{q,k}^T \hat{\Sigma}_{q,k}^{-1} \hat{\mathbf{H}}_{q,k}. \quad (30)$$

Here, $\hat{\mathbf{H}}_{q,k}$ and $\hat{\Sigma}_{q,k}$ denote the Jacobian and the measurement covariance matrix evaluated at $\xi_{k|k-1}^q$, respectively, where $\xi_{k|k-1}^q$ denotes the predicted state vector of the q th target for the case of zero process noise [23].

IV. SIMULTANEOUS MULTIBEAM RESOURCE ALLOCATION STRATEGY

A. Basis of the Technique

Mathematically, the SMRA strategy can be described as a problem of optimizing a certain system level utility function subject to some beam and power constraints. The BIM [16] derived in the previous section, bounds the error variance of the unbiased estimates of the unknown target state, and thus can be utilized as the optimization criterion for the resource allocation strategy. At every instant, we can analyze a set of $\mathbf{u}_k = [u_{1,k}, \dots, u_{Q,k}]^T$ and $\mathbf{P}_k = [P_{1,k}, \dots, P_{Q,k}]^T$ (candidate), and in each case determine a predictive BIM, from which we obtain the performance bounds.

We are then in a position to design and control limited radar resources to achieve the best performance. The general resource allocation strategy can be detailed as follows.

B. Predictive BCRLB and the Performance Metric

The crucial feature of the resource allocation scheme is that it must be predictive. At every instant, we use the information available to extrapolate the targets' positions in the future, and then, we organize the transmit parameters of the system to follow-up on the targets' paths. Given the updated BIM of the q th target $\mathbf{J}(\xi_{k-1}^q)$ at time index $k-1$ and a pair of candidate $(u_{q,k}, P_{q,k})$, we can now calculate the predictive BIM $\mathbf{J}(u_{q,k}, P_{q,k})|_{\xi_k^q}$ at time index k by using the Riccati-like recursion given in (30):

$$\mathbf{J}(u_{q,k}, P_{q,k})|_{\xi_k^q} = \mathbf{J}_P(\xi_k^q) + u_{q,k} \hat{\mathbf{H}}_{q,k}^T \hat{\Sigma}_{q,k}^{-1} \hat{\mathbf{H}}_{q,k}. \quad (31)$$

Then, the predictive BCRLB of the q th target is defined as the inverse of the BIM:

$$\mathbf{C}_{CRLB}(u_{q,k}, P_{q,k}) = \left[\mathbf{J}(u_{q,k}, P_{q,k}) | \boldsymbol{\xi}_k^q \right]^{-1}. \quad (32)$$

The diagonal elements of $\mathbf{C}_{CRLB}(u_{q,k}, P_{q,k})$ denote the lower bound on the estimation MSE of the q th target's state.

To accomplish the resource allocation strategy, it is sufficient for us to utilize the worst case tracking BCRLB as a criterion for the SMRA strategy

$$\mathbb{F}(\mathbf{u}_k, \mathbf{P}_k) = \max_q \sqrt{\text{Tr}[\mathbf{C}_{CRLB}(u_{q,k}, P_{q,k})]} \quad (33)$$

where (33) can be used to gauge the worst case tracking performance. Herein, the expression 'worst case' actually denotes the target with a maximum BCRLB. It is worthwhile noting that we assume that one beam can only track a single target for simplicity in this paper. However, in practice, there could exist a case in which multiple targets are covered by a single beam. Fortunately, we show that the SMRA can be easily extended to this case by replacing the criterion $\mathbb{F}(\mathbf{u}_k, \mathbf{P}_k)$ in Appendix A.

C. Criterion Minimization for a Predetermined Beam and Power Resource

According to (31), we know that the target tracking accuracy is related to many parameters. The adaptable parameters that are considered in this paper are the binary vector $\mathbf{u}_k$ (decide how many and which targets to be tracked) and the transmit power vector $\mathbf{P}_k$. For the predetermined maximum beam number M and the total power budget P_{total} at each time index, the aim of our work is to optimally allocate the limited beam and power resources, which can result in the minimization of the predicted worst case BCRLB, subject to the power limit of each beam. The resulting optimization problem is

$$\begin{aligned} & \min_{P_{q,k}, q=1, \dots, Q} [\mathbb{F}(\mathbf{u}_k, \mathbf{P}_k)] \\ \text{s.t.} \quad & \begin{cases} P_{q,k} = 0 & u_{q,k} = 0 \\ P_{q,k} \geq \bar{P}_{\min} & u_{q,k} = 1 \end{cases} \\ & u_{q,k} \in \{0, 1\} \\ & \mathbf{1}_Q^T \mathbf{P}_k = P_{\text{total}}, \quad \mathbf{1}_Q^T \mathbf{u}_k \leq M, \end{aligned} \quad (34)$$

where $\mathbf{1}_Q^T = [1, 1, \dots, 1]_{1 \times Q}$, and the transmit power of the q th beam is constrained by a minimum value $\bar{P}_{\min}$ if $u_{q,k} = 1$.

The optimization problem in (34) is nonconvex because of the binary vector $\mathbf{u}_k$. To solve this problem, a common trick is to partition the two optimization variables $(\mathbf{u}_k, \mathbf{P}_k)$. Before proceeding to solve problem (34), we provide the following descriptions and some necessary lemmas, whose proofs are found in Appendix B.

At every instant, the maximum beam number is M . Therefore, the feasible beam number at time index k is $m_k \in \{1, 2, \dots, M\}$. In this case, for a different m_k , the corresponding beam allocation vector can be represented as $\mathbf{u}_k^{m_k}$, where $\mathbf{1}_Q^T \mathbf{u}_k^{m_k} = m_k$.

Lemma 1: For a given m_k , the optimal solution to the binary variable $\mathbf{u}_{k,\text{opt}}^{m_k}$ can be uniquely determined regardless of $\mathbf{P}_k$.

Lemma 2: For a given $\mathbf{u}_{k,\text{opt}}^{m_k}$, the power allocation problem can be reformulated as a convex problem.

By making use of lemma 1, we first calculate $\mathbf{u}_{k,\text{opt}}^{m_k}$ as

$$\mathbf{u}_{k,\text{opt}}^{m_k}(i) = \begin{cases} 1 & \text{if } i \in \mathbf{IX}(1 : m_k) \\ 0 & \text{else} \end{cases}. \quad (35)$$

where the definition of $\mathbf{IX}$ is given in (51). Then, enumerating all of the $\mathbf{u}_{k,\text{opt}}^{m_k}$, $m_k = 1, \dots, M$, problem (34) can be formed as

$$\begin{aligned} & \min_{m_k=1, \dots, M} \left\{ \min_{P_{q,k}, q=1, \dots, Q} [\mathbb{F}(\mathbf{u}_{k,\text{opt}}^{m_k}, \mathbf{P}_k^{m_k})] \right\} \\ \text{s.t.} \quad & \begin{cases} P_{q,k} = 0 & u_{q,k}^{m_k} = 0 \\ P_{q,k} \geq \bar{P}_{\min} & u_{q,k}^{m_k} = 1 \end{cases} \\ & \mathbf{1}_Q^T \mathbf{P}_k^{m_k} = P_{\text{total}}, \end{aligned} \quad (36)$$

For a specified $\mathbf{u}_{k,\text{opt}}^{m_k}$, the corresponding optimal power allocation results $\mathbf{P}_{k,\text{opt}}^{m_k}$ can be obtained by solving

$$\begin{aligned} & \min_{P_{q,k}, q=1, \dots, Q} [\mathbb{F}(\mathbf{u}_{k,\text{opt}}^{m_k}, \mathbf{P}_k^{m_k})] \\ \text{s.t.} \quad & \begin{cases} P_{q,k} = 0 & u_{q,k}^{m_k} = 0 \\ P_{q,k} \geq \bar{P}_{\min} & u_{q,k}^{m_k} = 1 \end{cases} \\ & \mathbf{1}_Q^T \mathbf{P}_k^{m_k} = P_{\text{total}}. \end{aligned} \quad (37)$$

Defining a new power vector $\tilde{\mathbf{P}}_k^{m_k} = \mathbf{P}_k^{m_k} [\mathbf{IX}(1 : m_k)]$ of length m_k , problem (37) can be reformulated as

$$\begin{aligned} & \min_{\tilde{\mathbf{P}}_k^{m_k}} [\mathbb{G}(\tilde{\mathbf{P}}_k^{m_k})] \\ \text{s.t.} \quad & P_{\mathbf{IX}(i),k} \geq \bar{P}_{\min}, \quad i = 1, \dots, m_k \\ & \mathbf{1}_{m_k}^T \tilde{\mathbf{P}}_k^{m_k} = P_{\text{total}} \end{aligned} \quad (38)$$

where

$$\begin{aligned} & \mathbb{G}(\tilde{\mathbf{P}}_k^{m_k}) \\ &= \max_{q \in \mathbf{IX}(1 : m_k)} \left\{ \text{Tr} \left(\left[\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \hat{\mathbf{H}}_{q,k}^T \boldsymbol{\Sigma}_{q,k}^{-1} (P_{q,k}) \hat{\mathbf{H}}_{q,k} \right]^{-1} \right) \right\}. \end{aligned} \quad (39)$$

Lemma 2 shows that (39) is a convex problem. The results in [27] suggest that the *gradient projection* (GP) method can be extremely effective in solving multidimensional problems that have many simple constraints, such as lower and/or upper bounds on the variables. Therefore, the optimal power vector can be efficiently obtained by the GP algorithm presented in Table I, such that resource-aware technique can be utilized in real-time systems. In Fig. 5, the sketch map of the GP algorithm is given, which can help us have insight into this method.

In this case, the optimal power allocation result $\mathbf{P}_{k,\text{opt}}^{m_k}$ for a determined $\mathbf{u}_{k,\text{opt}}^{m_k}$ can be calculated as

$$\begin{cases} \mathbf{P}_{k,\text{opt}}^{m_k} [\mathbf{IX}(1 : m_k)] = \tilde{\mathbf{P}}_{k,\text{opt}}^{m_k} \\ \mathbf{P}_{k,\text{opt}}^{m_k} [\mathbf{IX}(m_k + 1 : Q)] = \mathbf{0} \end{cases}. \quad (40)$$

Finally, by solving a set of M convex optimization problems, the optimal solution $(\mathbf{u}_{k,\text{opt}}, \mathbf{P}_{k,\text{opt}})$ can be achieved by

$$(\mathbf{u}_{k,\text{opt}}, \mathbf{P}_{k,\text{opt}}) = \arg \min_{\mathbf{u}_{k,\text{opt}}^{m_k}, \mathbf{P}_{k,\text{opt}}^{m_k}, m_k=1, \dots, M} [\mathbb{F}(\mathbf{u}_{k,\text{opt}}^{m_k}, \mathbf{P}_{k,\text{opt}}^{m_k})]. \quad (41)$$

TABLE I
THE GP ALGORITHM FOR POWER ALLOCATION

- (1) Setting $l=1$, choose a feasible point $\tilde{\mathbf{P}}_{k,l}^{m_k} = P_{\text{total}}/m_k \cdot \mathbf{1}_{m_k}$;
- (2) Denote the constraints of (38) into vector form and separate them into two parts: $\mathbf{A}_1 \tilde{\mathbf{P}}_{k,l}^{m_k} = \mathbf{b}_1$ and $\mathbf{A}_2 \tilde{\mathbf{P}}_{k,l}^{m_k} > \mathbf{b}_2$. The general active constraint is $\mathbf{A}_a \tilde{\mathbf{P}}_{k,l}^{m_k} = (\mathbf{A}_1^T, \mathbf{1}_{m_k}^T)^T \cdot \tilde{\mathbf{P}}_{k,l}^{m_k} = (\mathbf{b}_1^T, P_{\text{total}})^T$;
- (3) Define the projection matrix $\mathbf{Pr} = \mathbf{I}_{m_k} - \mathbf{A}_a^T (\mathbf{A}_a \mathbf{A}_a^T)^{-1} \mathbf{A}_a$;
- (4) Let $\mathbf{d}_{k,l} = -\mathbf{Pr} \cdot \Delta_{\tilde{\mathbf{P}}_{k,l}^{m_k}} \mathbb{G}(\tilde{\mathbf{P}}_{k,l}^{m_k})$. If $\mathbf{d}_{k,l} = \mathbf{0}_{m_k \times 1}$, then go to (6); otherwise, go to (5);

$$(5) \text{ Let } \bar{\lambda} = \begin{cases} \min \left\{ \frac{\left| \mathbf{a}_i \tilde{\mathbf{P}}_{k,l}^{m_k} - (\mathbf{b}_2)_i \right|}{\mathbf{a}_i \mathbf{d}_{k,l}} \mid \mathbf{a}_i \mathbf{d}_{k,l} < 0, \mathbf{a}_i \in \mathbf{A}_2 \right\} \\ \mathbf{A}_2 \mathbf{d}_{k,l} \geq 0 \\ +\infty \quad \mathbf{A}_2 \mathbf{d}_{k,l} \geq 0 \end{cases},$$

determine the step length $\lambda_{k,l} : \min_{0 \leq \lambda \leq \bar{\lambda}} [\mathbb{G}(\tilde{\mathbf{P}}_{k,l}^{m_k} + \lambda \mathbf{d}_{k,l})] = \mathbb{G}(\tilde{\mathbf{P}}_{k,l}^{m_k} + \lambda_{k,l} \mathbf{d}_{k,l})$

Then, we have $\tilde{\mathbf{P}}_{k,l+1}^{m_k} = \tilde{\mathbf{P}}_{k,l}^{m_k} + \lambda_{k,l} \mathbf{d}_{k,l}$, $l = l+1$, and go to (2);

- (6) Calculate $(\mathbf{A}_a \mathbf{A}_a^T)^{-1} \mathbf{A}_a \cdot \Delta_{\tilde{\mathbf{P}}_{k,l}^{m_k}} \mathbb{G}(\tilde{\mathbf{P}}_{k,l}^{m_k}) = (\mathbf{v}_1^T, \mathbf{v}_2^T)^T$, and let $\mathbf{v}_j = \min(\mathbf{v}_j)$. If $\mathbf{v}_j > 0$, then stop and take $\tilde{\mathbf{P}}_{k,\text{opt}}^{m_k} = \tilde{\mathbf{P}}_{k,l}^{m_k}$; otherwise, drop the $\mathbf{a}_j$ that is associated with the negative multiplier $\mathbf{v}_j$ from the active set $\mathbf{A}_a$, and then, go to (3).

TABLE II
THE GENERAL STEPS OF THE BAYESIAN TRACKER WITH SMRA

- (1) Let $k=2$, and assume an initial PDF $p(\xi^q)$, $q=1,2,\dots,Q$ and the particle number L ; set $\mathbf{P}_{k,\text{opt}} = [P_{\text{total}}/M, \mathbf{1}_M^T, \mathbf{0}_{Q-M}^T]^T$;
- (2) Draw L samples according to $p(\xi^q | \xi_{k-1}^q)$, to obtain $\{\xi_{k-1}^q\}_{i=1}^L$;
- (3) For $q=1,2,\dots,Q$
 - (3.1) Give the measurements $\mathbf{z}_{q,k}(u_{q,k}, P_{q,k})$, and define $\omega_k^{qi} = p(\mathbf{z}_{q,k} | \xi_k^{qi})$, $i=1,2,\dots,L$;
 - (3.2) Perform normalization: $\omega_k^{qi} = \omega_k^{qi} / \sum_{i=1}^L \omega_k^{qi}$;
 - (3.3) Resample L times from $\{\xi_k^{qi}, \omega_k^{qi}\}_{i=1}^L$ and achieve $\{\xi_k^{qi}, 1/L\}_{i=1}^L$.
The basic idea of resampling is to eliminate particles that have small weights and concentrate on particles that have large weights. Here, we adopt the systematic resampling technique mentioned in reference [29].
 - (3.4) Compute estimates based on these samples and weights $\hat{\xi}_k^q \approx \sum_{i=1}^L (1/L \cdot \xi_k^{qi})$;
- (4) Calculate the criterion $\mathbb{F}(\mathbf{u}_{k+1}, \mathbf{P}_{k+1})$ according to (31)-(33) and implement the SMRA strategy detailed in Section IV-C;
- (5) Send the allocation result $\mathbf{u}_{k+1,\text{opt}}$ and $\mathbf{P}_{k+1,\text{opt}}$ to the radar system to form the illumination strategy;
- (6) Let $k=k+1$, propagate $\{\xi_{k-1}^q\}_{i=1}^L$ according to $\xi_k^{qi} = \mathbf{F}_{k-1}^q \xi_{k-1}^{qi} + \boldsymbol{\eta}_{k-1}^{qi}$, and go to (3).

D. Target State Estimation

In this paper, we adopt a centralized tracking method with an indirect technique [28] to estimate the target state. With indirect estimation, the range, bearing, Doppler shift and RCS of each target are estimated first. The advantage of this method is

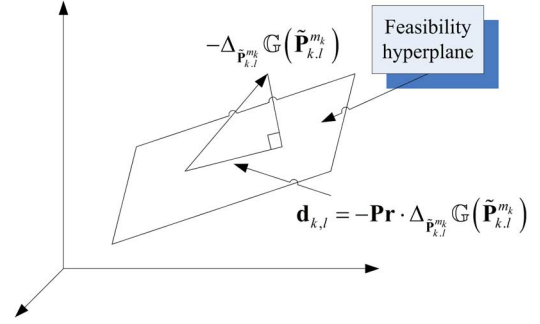


Fig. 5. The sketch map of the GP method.

that the indirect estimation technique offers data compression. In this case, we must achieve the target state by adopting the model given in (6) and (12).

It has been shown in [20] that the more nonlinear the model is, the higher potential the SIR-PF has. In our model, the measurement equation is highly nonlinear. Hence, we adopt SIR-PF to achieve an estimate of the hybrid state vector. Reference [20] indicates that the number of particles L must be chosen to be quite large, to increase the robustness. On the other hand, previous work in [23] also shows that the complexity of PF grows linearly with L . Therefore, we must choose a moderate particle number L to achieve a much better tradeoff between the performance and the complexity.

Overall, the closed-loop tracking system can be described as follows. First, the state of each target at the current time index is achieved by SIR-PF. Then, the predicted information (predictive BIM) is fed back, based on which resource allocation strategy is implemented. Finally, the allocation results are sent back to guide the probing strategy at the next time index, thereby rendering it a closed-loop system.

V. SIMULATION RESULTS

To illustrate the effectiveness of the proposed algorithm, we present some numerical results in this section. The worst case tracking accuracy achieved with the proposed resource allocation strategy is used as a metric to compare with the benchmark. Here, the tracking accuracy, obtained by $(\mathbf{u}_{k,\text{opt}}^{m_k}, \mathbf{P}_0^{m_k})$, $m_k = 1, \dots, M$, will be used as the benchmark, where $\mathbf{P}_0^{m_k}$ is defined as

$$\begin{cases} \mathbf{P}_0^{m_k} [\mathbf{IX}(1 : m_k)] = \frac{P_{\text{total}}}{m_k} \\ \mathbf{P}_0^{m_k} [\mathbf{IX}(m_k + 1 : Q)] = 0 \end{cases} \quad (42)$$

The benchmark is utilized to represent the conventional simultaneous working mode, where the number of beams is fixed at m_k and the power resource is uniformly allocated at each time index.

Then, to better reveal the effects of several factors (the target spread, target RCS and accuracy of the dynamic model) on the resource allocation results, we consider two RCS models

- (1) $\mathbf{H}_1: \mathbf{h}^q = [h_{q,1}, \dots, h_{q,k}]^T = [1, 1, \dots, 1]^T$, $q=1, 2, \dots, Q$;
- (2) $\mathbf{H}_2: \mathbf{h}^q = [h_{q,1}, \dots, h_{q,k}]^T = [1, 1, \dots, 1]^T$, $q \neq 1, 2, 4, 9$,

TABLE III
 THE PARAMETERS OF EACH TARGET

Target index	1	2	3	4	5	6	7	8	9
Position (km)	(3,55)	(-43,45)	(233,60)	(280,50)	(52,80)	(-35,15)	(200,80)	(208,10)	(100,66)
Distance (km)	126	164	137	176	106	149	123	96	74
Velocity (m/s)	(300,0)	(100,-150)	(200,-200)	(10,-200)	(200,100)	(100,150)	(100,-100)	(10,-200)	(200,-30)

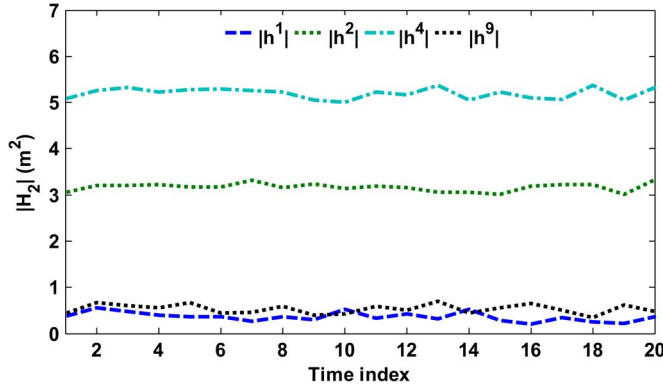
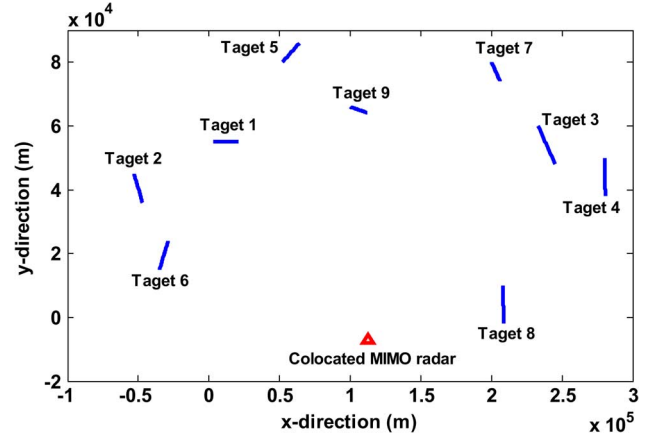

 Fig. 6. The RCS of the remaining targets in H_2 .


Fig. 7. Deployment of the targets with respect to radar.

and the RCS of the remaining targets are shown in Fig. 6, and two dynamic models

$$(1) \mathbf{S}_1 : s_q = 100, q = 1, 2, \dots, Q; (2) \mathbf{S}_2 : s_q = \begin{cases} 10000 & \text{if } q = 8 \\ 0.5 & \text{if } q = 6; \\ 100 & \text{others} \end{cases}$$

Combining these different target RCS models and target motion models, we investigate the following three cases. In each case, our colocated MIMO radar system is located at (112.75, -7) km. We assume that the number of radiating elements is $N = 32$, and thus, the beamwidth of each beam is set to $\theta \approx 100/N^\circ$ [25]. The signal's effective bandwidth and the effective time duration of the q th beam are set to $\beta_{q,k} = 1$ MHz and $T_{q,k} = 1$ ms, respectively. The carrier frequency of each beam is set to 1 GHz. The time interval between successive frames is set to $T_0 = 3$ s, and a sequence of 20 frames of data are utilized to support the simulation. The lower bound of the power of each beam is set to $\bar{P}_{\min} = 0.05P_{\text{total}}$. The number of particles is set to $L = 200$, as using more particles gives no noticeable improvement after convergence. The number of targets is set to $Q = 9$, and the parameters of each target are given in Table III.

The angular spread of these targets with respect to our radar system is given in Fig. 7.

A. Case 1: $\mathbf{H}_1$ and $\mathbf{S}_1$

This case supports the evaluation of the resource allocation strategy with the target RCS and the level of the process noise factored out. Therefore, the allocation of the beam and power resource compensate only for the deployment of the targets.

The worst case tracking BCRLB, which is achieved with the benchmark parameters $(\mathbf{u}_{k,\text{opt}}^{m_k}, \mathbf{P}_{k,\text{opt}}^{m_k})$ and the optimal parameter $(\mathbf{u}_{k,\text{opt}}, \mathbf{P}_{k,\text{opt}})$, are given in Fig. 8. To justify the utilization of the tracking steps presented in Section IV-D, the worst case position BCRLB, achieved with $(\mathbf{u}_{k,\text{opt}}, \mathbf{P}_{k,\text{opt}})$, and the corresponding tracking *root MSE* (RMSE), are shown in the subfigure of Fig. 8. Here, the position RMSE at the k th tracking interval is defined as

$$\text{RMSE}_k = \max_q \left[\sqrt{\frac{1}{N_{MC}} \sum_{j=1}^{N_{MC}} \left[\left(x_{Tk}^q - \hat{x}_{Tk}^{q,j} \right)^2 + \left(y_{Tk}^q - \hat{y}_{Tk}^{q,j} \right)^2 \right]} \right]. \quad (43)$$

where N_{MC} is the number of Monte Carlo trials. $(\hat{x}_{Tk}^{q,j}, \hat{y}_{Tk}^{q,j})$ is the state estimate of the q th target at the j th trial.

In Fig. 8, the initial BIMs $\mathbf{J}(\boldsymbol{\xi}_1^q), q = 1, 2, \dots, Q$, as well as the BCRLBs of each target are the same. During the first few moments, the worst case BCRLB increases with time. The reason is that the worst case BCRLBs at those times are from the targets, which have not been illuminated. (The prediction information on these targets is utilized to achieve the state estimate.) Intuitively speaking, the first turning point of the line in Fig. 8 appears when all of the targets are illuminated at least once. Hence, the larger the number of beams m_k , the faster the first turning point will occur. Overall, the benchmarks exhibit a higher BCRLB compared with an optimized distribution of the

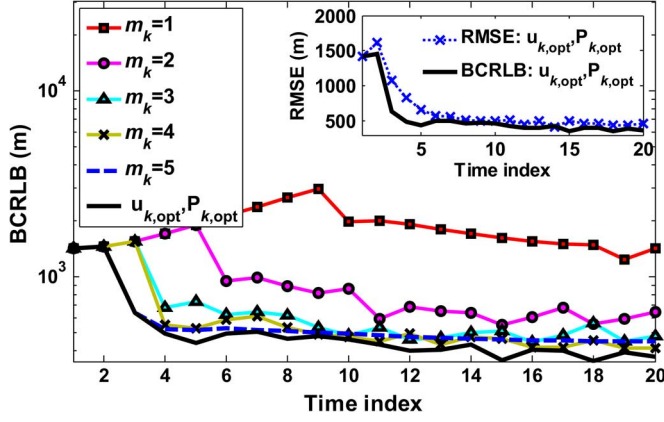


Fig. 8. The worst case BCRLB of the multiple targets in case 1.

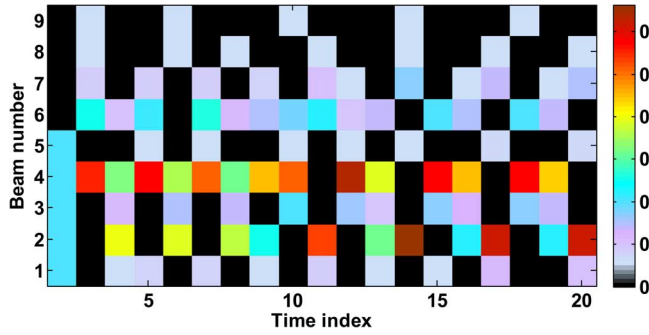


Fig. 9. The allocation results of the SMRA strategy in case 1.

beam and power resource. In other words, the SMRA method dilutes the error variation through having an adequate distribution of the beam and power resource. In our simulation, the initial locations of different targets are set according to the initial BIM $\mathbf{J}(\xi_1^q)$, $q = 1, 2, \dots, Q$, and thus, the initial RMSEs are consistent with the corresponding BCRLBs. The results in the subfigure imply that the tracking accuracy approaches the BCRLB as the number of measurements increases, and this finding verifies the correctness of the SMRA strategy.

To disclose the effects of the deployment of the targets on the allocation results, Fig. 9 depicts the beam and power allocation results averaged over 100 Monte Carlo trials. In this figure, the black areas at each time index indicate that $u_{q,k} = 0$, while others imply that $u_{q,k} = 1$, with different colors denoting the ratio of the transmit power. Herein, the ratio of the transmit power is defined as

$$r_{q,k} = \frac{P_{q,k}}{P_{\text{total}}}. \quad (44)$$

At time index $k = 2$, the predicted information of different targets is the same as the initial BIMs $\mathbf{J}(\xi_1^q)$, $q = 1, 2, \dots, Q$, and the covariances of the process noise $\mathbf{Y}_{\xi,k-1}^q$ are the same as well. Therefore, the number of beams is set to M , and the power resource is uniformly allocated to arbitrary M beams. For the other time index, more beam and power resources are allocated to beam 2 and beam 4, as increasing the transmit power of the beams that point to the targets that are farther from the radar will have a stronger impact on the worst case tracking performance. In contrast, less resource is distributed to beam 8 and beam 9, as these two targets are closer to our radar system. Then, we

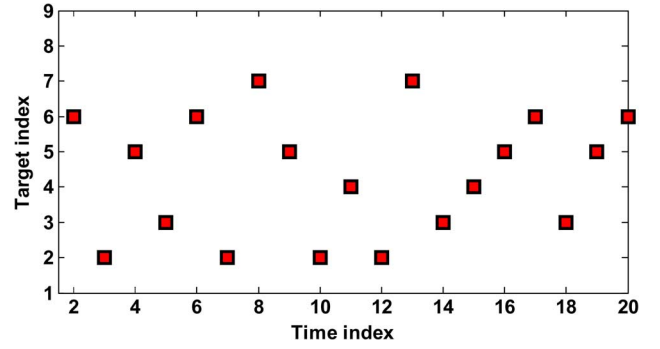


Fig. 10. The indices of the target with the maximum BCRLB for case 1.

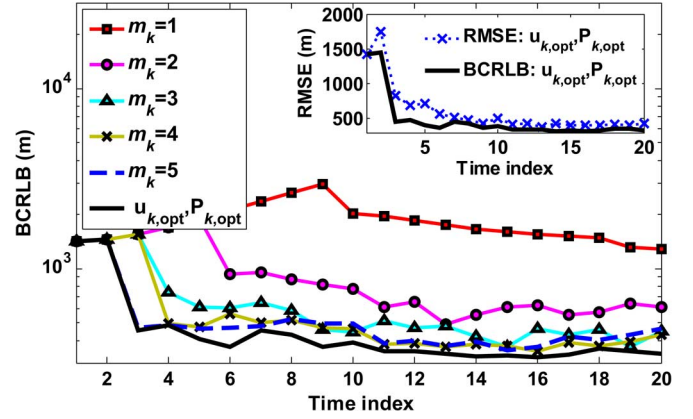


Fig. 11. The worst case BCRLB of the multiple targets in case 2.

conclude that the power will be added to the targets that are farther away from the radar.

At each time index, the worst case result could come from a different target. In Fig. 10, the indices of the target with the maximum BCRLB are shown for case 1. Combining the results in Fig. 9, we know that the targets with the largest BCRLB at time index k , which are shown in Fig. 10, will be illuminated at time index $k + 1$.

B. Case 2: $\mathbf{H}_2$ and $\mathbf{S}_1$

We then expand our simulation while considering the losses due to the target RCS. The worst case tracking performance is evaluated in Fig. 11 for the second case. The results prove that the SMRA algorithm can make more complete use of the limited resources of the radar system. For case 2, the results basically show that using 4 or 5 beams is better, when the power is uniformly distributed. The results in the subfigure imply that the tracking accuracy approaches the BCRLB as the number of measurements increases.

Fig. 12 illustrates the effect of the target RCS on the allocation results. In this case, two targets with lower reflectivity (target 1 and target 9) are allocated more beam and power resources, compared with the allocation results of case 1. Moreover, target 4 in this case is distributed with a lower level of resources compared with case 1, as it has a higher reflectivity. The results show that the resulting transmit power vector $\mathbf{P}_{k,\text{opt}}$ allocates most of the energy to the targets that have smaller reflectivity.

In Fig. 12, m_k is equal to the number of grids with different colors in the k th column. Hence, the results show that using 5 beams is better in case 2 (except for $k = 9$), when the power is

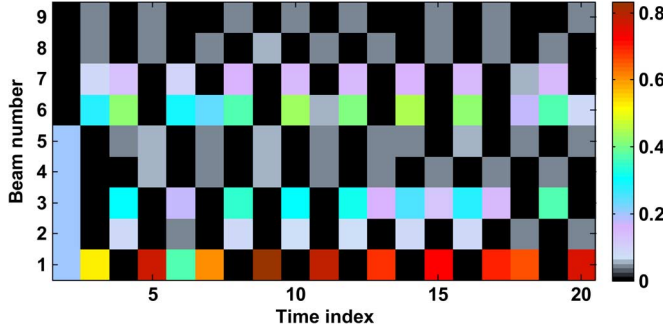


Fig. 12. The allocation results of the SMRA strategy in case 2.

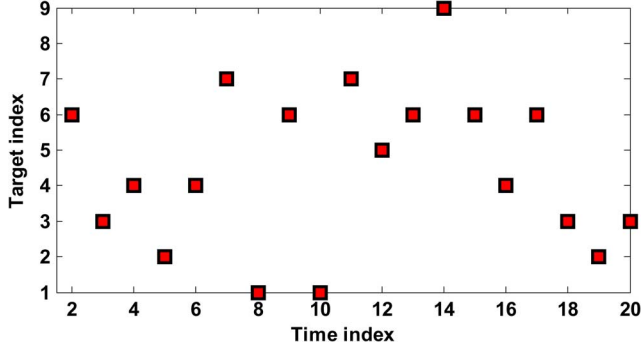


Fig. 13. The indices of the target with the maximum BCRLB for case 2.

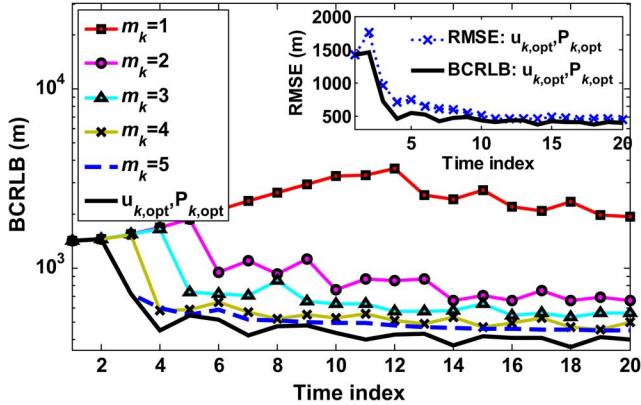


Fig. 14. The worst case BCRLB of the multiple targets in case 3.

optimally distributed. In Fig. 13, the indices of the target with the largest BCRLB are listed for case 2. Similarly, the target marked in Fig. 13 will be illuminated at the subsequent time index.

C. Case 3: $\mathbf{H}_1$ and $\mathbf{S}_2$

In this part, we analyze the effect of the target dynamic model on the allocation results. In this case, the dynamic model of target 8 is inaccurate. The worst case tracking performance is given in Fig. 14 for case 3, according to which we know the superiority of the proposed SMRA strategy.

The relationship between the accuracy of the dynamic model and the allocation results is highlighted in Fig. 15. At time index $k = 2$, the predicted information of target 8 is larger, as it has an inaccurate dynamic model. Hence, more beams are allocated to target 8, as it has an inaccurate dynamic model. In contrast, fewer beams and relatively less power is allocated to target 6, which has a precise dynamic model. The results yield a common

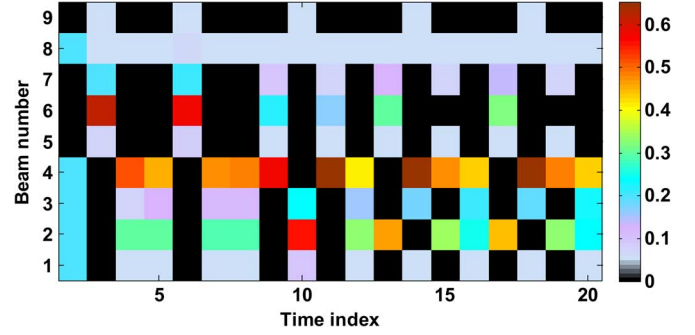


Fig. 15. The allocation results of the SMRA strategy in case 3.

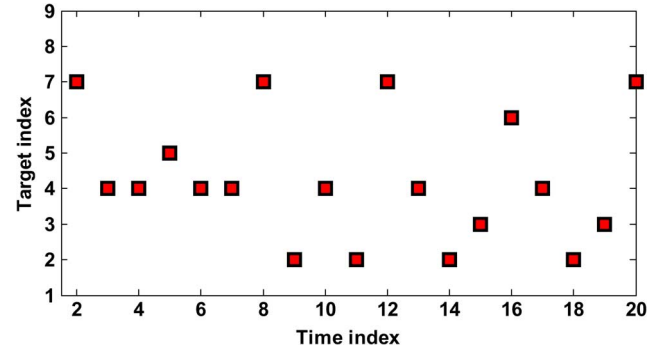


Fig. 16. The indices of the target with the maximum BCRLB for case 3.

finding that the resource will be added to the targets that have a less accurate dynamic model, in such a way that the worst case tracking accuracy will be more improved. Physically speaking, the ellipse in Fig. 4 denotes the prior track information of each target, which is related to the accuracy of the dynamic model. The sector area in Fig. 4(b) approximates the spatial variance distribution of the target measurement error. In this scenario, the intersection area actually denotes the tracking accuracy of each target. In case 3, the ellipse of target 8 is larger than that in case 1, and thus, we must allocate more resources to target 8, with the aim of reducing the area of the intersecting part.

The indices of the target with the largest BCRLB are given in Fig. 16 for case 3. Although target 4 in this case is assigned increased resources in terms of the amount of power and the number of beams, it is still the target that has the worst tracking performance often.

In Table IV, the tracking BCRLBs of all of the targets in case 3 are given, which can help us have insight into the resource allocation strategy. Herein, $\mathbf{C}_{CRLB}^P(\xi_k^q)$ denotes the prior CRLB of the q th target defined in (49), while $\mathbf{C}_{CRLB}(\xi_k^q)$ is the tracking BCRLB illustrated in (32).

Combining the results in Fig. 14, Fig. 15 and Table IV, we can conclude the following:

- 1) At each time index, the prior BCRLB of target 8 is higher, as the level of the process noise of this target is large;
- 2) At each time index, the beam resource is allocated to the targets with larger $\text{Tr}[\mathbf{C}_{CRLB}^P(\xi_k^q)]$ (the targets' indices correspond to the red underlined values in Table IV);
- 3) At each time index, power resources are also distributed to the targets that have a larger $\text{Tr}[\mathbf{C}_{CRLB}(\xi_k^q)]$. Among these targets, a larger amount of power is allocated to the targets that have a relatively larger distance or/and lower reflectivity conditions;

TABLE IV
THE TRACKING BCRLB OF EACH TARGET AT EVERY INSTANT

		1	2	3	4	5	6	7	8	9	Worst BCRLB
$k = 2$	$\text{Tr}[\mathbf{C}_{CRLB}^P(\xi_k^q)]$	1450	1450	1450	1450	1450	1449	1450	1530	1450	
	$\text{Tr}[\mathbf{C}_{CRLB}(\xi_k^q)]$	305	573	382	659	1450	1449	1450	226	1450	1450
$k = 3$	$\text{Tr}[\mathbf{C}_{CRLB}^P(\xi_k^q)]$	386	637	454	722	1545	1539	1545	764	1545	
	$\text{Tr}[\mathbf{C}_{CRLB}(\xi_k^q)]$	386	637	454	722	290	289	288	288	140	722
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$k = 20$	$\text{Tr}[\mathbf{C}_{CRLB}^P(\xi_k^q)]$	506	406	509	404	347	368	400	782	245	
	$\text{Tr}[\mathbf{C}_{CRLB}(\xi_k^q)]$	326	326	323	326	347	368	400	287	245	400

- 4) After the power allocation, the BCRLB of the targets, which do not have minimum power, are almost the same.

VI. CONCLUSIONS

A colocated MIMO radar system has the ability of addressing multiple beam information; thus, it has more advantages over traditional radar systems, such as a lower probability of interception and better system sensitivity. However, the simultaneous multibeam working mode has two finite working resources: the maximum beam number and the total transmit power of the multiple beams. Therefore, the scheme of SMRA is considered to minimize the worst case tracking BCRLB. It is shown that the corresponding resource allocation problem can be reformulated as a set of M convex problems. In this scenario, the optimal solution can easily be obtained to meet the demands of a real-time system. Simulation results show that the improvements in the worst target's tracking accuracy can be obtained through the proposed SMRA strategy. The results also show that targets are allocated more resources when they are farther from the radar, have less accurate dynamic models or/and have weaker reflectivity conditions.

APPENDIX A

In this Appendix, we derive the criterion $\mathbb{F}(\mathbf{u}_k, \mathbf{P}_k)$ for the scenario that multiple targets can be covered by a single beam. Assume that the indices of the targets, which are illuminated by the l th beam, are collected, as represented by $\Pi_l = (i, \dots, j)$. Then, given the transmit power $P_{l,k}$, we can achieve the maximum BCRLB of the l th beam

$$\begin{aligned} \psi_l(P_{l,k}) &= \begin{cases} \max_{c \in \Pi_l} (\text{Tr}[\mathbf{J}_P^{-1}(\xi_k^c)]) & \text{if } P_{l,k} = 0 \\ \max_{c \in \Pi_l} (\text{Tr}[\mathbf{J}^{-1}(u_{l,k} = 1, P_{l,k})|_{\xi_k^c}]) & \text{if } P_{l,k} > 0 \end{cases} \end{aligned} \quad (45)$$

where $\mathbf{J}(u_{l,k}, P_{l,k})|_{\xi_k^c}$ is defined in (31). Finally, the criterion is the maximum BCRLB among multiple targets

$$\mathbb{F}(\mathbf{u}_k, \mathbf{P}_k) = \max \left\{ \max_{l=1, \dots, M} [\psi_l(P_{l,k})], \max_{j \in \Omega} (\text{Tr}[\mathbf{J}_P^{-1}(\xi_k^j)]) \right\}. \quad (46)$$

where Ω denotes the set of un-illuminated targets, and we have

$$|\Pi_1 \cup \Pi_2 \dots \Pi_M \cup \Omega| = Q. \quad (47)$$

where $|\mathbf{X}|$ denotes the cardinality or number of elements in $\mathbf{X}$. With the optimization method detailed in Section IV-C, we can achieve the optimal allocation results for the scenario that one beam can detect more than one target.

APPENDIX B

Proof of lemma 1: Given the updated BIM of the q th target $\mathbf{J}(\xi_{k-1}^q)$ at time index $k-1$, the FIM of the prior information at instant k can be calculated as

$$\mathbf{J}_P(\xi_k^q) = \left(\mathbf{Y}_{\xi, k-1}^q + \mathbf{F}_{\xi}^q \mathbf{J}^{-1}(\xi_{k-1}^q) (\mathbf{F}_{\xi}^q)^T \right)^{-1}. \quad (48)$$

Clearly, if the initial BIM $\mathbf{J}(\xi_1^q) \in S_{++}$, then $\mathbf{J}_P(\xi_k^q) \in S_{++}$, where S_{++} denotes the set of symmetric positive definite matrices.

The inverse of the q th target's prior information is then

$$\mathbf{C}_{CRLB}^P(\xi_k^q) = \mathbf{J}_P^{-1}(\xi_k^q). \quad (49)$$

To determine $\mathbf{u}_k^{m_k}$, we collect $\mathbf{C}_{CRLB}^P(\xi_k^q)$ into a vector

$$\mathbf{C}^P = \left\{ \text{Tr}[\mathbf{C}_{CRLB}^P(\xi_k^1)], \dots, \text{Tr}[\mathbf{C}_{CRLB}^P(\xi_k^Q)] \right\}^T, \quad (50)$$

sort the elements of $\mathbf{C}^P$ and arrange them in descending order

$$[\mathbf{C}_{\text{order}}^P, \mathbf{IX}] = \text{sort}(\mathbf{C}^P). \quad (51)$$

where $\mathbf{IX}$ is a permutation vector of $\mathbf{C}^P$.

Next, we use the mathematical induction method to prove that for a determined beam number m_k , $\mathbf{u}_{k, \text{opt}}^{m_k}$ can be uniquely determined:

$$\mathbf{u}_{k, \text{opt}}^{m_k}(i) = \begin{cases} 1 & \text{if } i \in \mathbf{IX}(1 : m_k) \\ 0 & \text{else} \end{cases}. \quad (52)$$

Basis: Show that the statement holds for $m_k = 1$.

If $m_k = 1$, then the possible points can be enumerated as $[\mathbf{u}_{k, \mathbf{IX}(1)}^1, \dots, \mathbf{u}_{k, \mathbf{IX}(Q)}^1]$, where $\mathbf{u}_{k, q}^1$ denotes the event that the single beam is pointing to the q th target. The criterion is then shown in (53) at the top of the next page. We know that for an arbitrary allocation result $\mathbf{P}_{k, q}^1$ (which satisfies $\mathbf{P}_{k, q}^1(q) > 0$), the FIM of the data $\mathbf{H}_{q, k}^T \mathbf{\Sigma}_{q, k}^{-1}(\mathbf{P}_{k, q}^1(q)) \mathbf{H}_{q, k}$ is a positive definite matrix. Thus, according to (49), we have

$$\mathbf{C}_{\text{order}}^P(1) > \text{Tr} \left(\left[\mathbf{J}_P(\xi_k^q) + \hat{\mathbf{H}}_{q, k}^T \hat{\mathbf{\Sigma}}_{q, k}^{-1}(\mathbf{P}_{k, q}^1(q)) \hat{\mathbf{H}}_{q, k} \right]^{-1} \right) \quad (54)$$

for $q = \mathbf{IX}(1)$, and $\mathbf{C}_{\text{order}}^P(1) \geq \mathbf{C}_{\text{order}}^P(2)$. Above all, to minimize the criterion given in (53), the optimal solution must be

$$\mathbb{F}(\mathbf{u}_{k,q}^1, \mathbf{P}_{k,q}^1) = \begin{cases} \mathbf{C}_{\text{order}}^P(1) & \text{if } q \in \mathbf{IX}(2 : Q) \\ \max \left[\mathbf{C}_{\text{order}}^P(2), \text{Tr} \left(\left[\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \hat{\mathbf{H}}_{q,k}^T \hat{\boldsymbol{\Sigma}}_{q,k}^{-1} (\mathbf{P}_{k,q}^1(q)) \hat{\mathbf{H}}_{q,k} \right]^{-1} \right) \right] & \text{if } q = \mathbf{IX}(1) \end{cases} \quad (53)$$

$$\mathbb{F}(\mathbf{u}_{k,q}^{L+1}, \mathbf{P}_{k,q}^{L+1}) = \begin{cases} \max \left[\mathbf{C}_{\text{order}}^P(L+1), c_L \right] & \text{if } q \in \mathbf{IX}(L+2 : Q) \\ \max \left[\mathbf{C}_{\text{order}}^P(L+2), \text{Tr} \left(\left[\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \hat{\mathbf{H}}_{q,k}^T \hat{\boldsymbol{\Sigma}}_{q,k}^{-1} (\mathbf{P}_{k,q}^{L+1}(q)) \hat{\mathbf{H}}_{q,k} \right]^{-1} \right), c_L \right] & \text{if } q = \mathbf{IX}(L+1) \end{cases} \quad (57)$$

$\mathbf{u}_{k,\text{opt}}^1 = \mathbf{e}_{\mathbf{IX}(1)}^Q$. Thus, it has been shown that the statement holds for $m_k = 1$.

Inductive step: Show that if the statement holds for $m_k = L$, $1 < L < M$, then it also holds for $m_k = L + 1$. This goal can be accomplished as follows:

Assume that the statement holds (for some unspecified value of $m_k = L$); then, we have

$$\mathbf{u}_{k,\text{opt}}^L(i) = \begin{cases} 1 & \text{if } i \in \mathbf{IX}(1 : L) \\ 0 & \text{else} \end{cases} \quad (55)$$

In this case, the feasible points for $m_k = L + 1$ are $[\mathbf{u}_{k,\mathbf{IX}(L+1)}^{L+1}, \dots, \mathbf{u}_{k,\mathbf{IX}(Q)}^{L+1}]$. Herein, $\mathbf{u}_{k,q}^{L+1}$ is defined as

$$\mathbf{u}_{k,q}^{L+1}(i) = \begin{cases} 1 & \text{if } i \in \mathbf{IX}(1 : L) \parallel i = q \\ 0 & \text{else} \end{cases} \quad (56)$$

The criterion is then shown in (57) at the top of the page, where $\mathbf{P}_{k,q}^{L+1}$ is an arbitrary allocation result that satisfies $\mathbf{P}_{k,q}^{L+1}(q) > 0$, and

$$c_L = \max_{l \in \mathbf{IX}(1:L)} \text{Tr} \left(\left[\mathbf{J}_P(\boldsymbol{\xi}_k^l) + \hat{\mathbf{H}}_{l,k}^T \hat{\boldsymbol{\Sigma}}_{l,k}^{-1} (\mathbf{P}_{k,q}^{L+1}(l)) \hat{\mathbf{H}}_{l,k} \right]^{-1} \right). \quad (58)$$

Similarly, we also have

$$\mathbf{C}_{\text{order}}^P(L+1) > \text{Tr} \left(\left[\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \hat{\mathbf{H}}_{q,k}^T \hat{\boldsymbol{\Sigma}}_{q,k}^{-1} (\mathbf{P}_{k,q}^{L+1}(q)) \hat{\mathbf{H}}_{q,k} \right]^{-1} \right) \quad (59)$$

for $q = \mathbf{IX}(L+1)$, and $\mathbf{C}_{\text{order}}^P(L+1) \geq \mathbf{C}_{\text{order}}^P(L+2)$. Thus, to minimize the criterion $\mathbb{F}(\mathbf{u}_{k,q}^{L+1}, \mathbf{P}_{k,q}^{L+1})$, the optimal solution is

$$\mathbf{u}_{k,\text{opt}}^{L+1}(i) = \begin{cases} 1 & \text{if } i \in \mathbf{IX}(1 : L+1) \\ 0 & \text{else} \end{cases} \quad (60)$$

which shows that indeed the statement holds for $m_k = L + 1$.

Since both the basis and the inductive step have been proven, it has now been proven by mathematical induction that the statement holds for all $m_k \leq M$.

Proof of Lemma 2: The objective problem described in (37) can be reformulated as

$$\begin{aligned} & \min_{\tilde{\mathbf{P}}_k^{m_k}} \left[\mathbb{G}(\tilde{\mathbf{P}}_k^{m_k}) \right] \\ & \text{s.t. } P_{\mathbf{IX}(i),k} \geq \bar{P}_{\min}, \quad i = 1, \dots, m_k \\ & \mathbf{1}^T \tilde{\mathbf{P}}_k^{m_k} = P_{\text{total}} \end{aligned} \quad (61)$$

where

$$\begin{aligned} & \mathbb{G}(\tilde{\mathbf{P}}_k^{m_k}) \\ &= \max_{q \in \mathbf{IX}(1:m_k)} \left\{ \text{Tr} \left(\left[\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \hat{\mathbf{H}}_{q,k}^T \hat{\boldsymbol{\Sigma}}_{q,k}^{-1} (\mathbf{P}_{q,k}) \hat{\mathbf{H}}_{q,k} \right]^{-1} \right) \right\}. \end{aligned} \quad (62)$$

Let $\mathbf{U}_q = \text{Tr}([\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \hat{\mathbf{H}}_{q,k}^T \hat{\boldsymbol{\Sigma}}_{q,k}^{-1} (\mathbf{P}_{q,k}) \hat{\mathbf{H}}_{q,k}]^{-1})$, and we can reformulate it as

$$\mathbf{U}_q = \text{Tr} \left(\left[\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \sum_{i=1}^{n_z} (P_{q,k} a_{ii}) b_i^T b_i \right]^{-1} \right), \quad (63)$$

where b_i is the i th row of $\hat{\mathbf{H}}_{q,k}$. $(P_{q,k} a_{ii})$ is the i th diagonal element of $\hat{\boldsymbol{\Sigma}}_{q,k}^{-1} (\mathbf{P}_{q,k})$.

According to [30], we know that $\text{Tr}(\mathbf{X}^{-1})$ is convex on $\mathbf{X} \in S_{++}$. Because $\mathbf{U}_q$ is a composition of $\text{Tr}(\mathbf{X}^{-1})$ with an affine transformation $\mathbf{X} \rightarrow \mathbf{J}_P(\boldsymbol{\xi}_k^q) + \sum_{i=1}^{n_z} [(P_{q,k} a_{ii}) b_i^T b_i]$, we know that $\mathbf{U}_q$ is convex on $\{\mathbf{P}_k | [\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \sum_{i=1}^{n_z} [(P_{q,k} a_{ii}) b_i^T b_i]] \in S_{++}\}$ [30] (It is easy for us to prove that $\mathbf{J}_P(\boldsymbol{\xi}_k^q) + \sum_{i=1}^{n_z} [(P_{q,k} a_{ii}) b_i^T b_i]$ is a symmetric positive definite matrix, if the initial BIM $\mathbf{J}(\boldsymbol{\xi}_1) \in S_{++}$). Then, we know that the pointwise maximum $\mathbb{G}(\tilde{\mathbf{P}}_k^{m_k})$ of $\mathbf{U}_{\mathbf{IX}(1)}, \dots, \mathbf{U}_{\mathbf{IX}(m_k)}$ is also convex.

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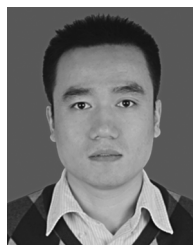
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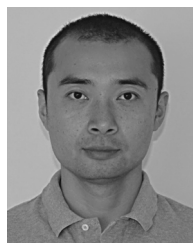
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